PEIJING PANG

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EDUCATION

Oct. 2025 - Jun. 2026 **University of Cambridge**
MPhil in Data Intensive Science (Pass)
Relevant modules: Deep Learning (78.6%), Applications of Data Science to Cosmology (75.4%), Machine Learning (75.4%), Applications of Data Science to High Energy Physics (72.0%), Statistical Methods (70.7%)

Sep. 2021 - Jun. 2025 **University College London**
MSci in Physics, First-Class Honours
Relevant modules: Advanced Physical Cosmology (95.7%), Particle Physics (83.2%), Galaxy Dynamics, Formation and Evolution (82.3%), Advanced Quantum Theory (79.2%), Practical Machine Learning for Physicists (78.2%)

RESEARCH EXPERIENCE

Jan. 2026 - Jun. 2026 **Transformer-Based Jet-Flavour Tagging for FCC-ee**
MPhil research project | University of Cambridge | Supervisor: Dr Matthew Monk
- Built a ROOT-to-PyTorch pipeline for 1.2 million simulated jets and adapted a DeepJetTransformer-style baseline to classify six jet flavours.
- Implemented a light-flavour specialist with pairwise attention bias and fused its predictions with the baseline, improving test accuracy from 61.28% to 62.48% on 180,000 simulated jets.
- Improved u-versus-d signal efficiency from 14.72% to 20.43% at 10% background efficiency; feature ablations identified charge-pair information as a key contributor.
- Ran GPU training with Slurm and compared hyperparameters, feature sets and performance across jet kinematics, examining trade-offs between light-flavour classes.

Jul. 2025 - Sep. 2025 **UV Spectroscopy for Trace Krypton Detection in Xenon for the PandaX Experiment**
Research Intern | Tsung-Dao Lee Institute, Shanghai Jiao Tong University
Supervisors: Professor Jianglai Liu and Dr Jiaqi Hui
- Contributed to experimental design, apparatus selection and procurement for UV spectroscopy of trace krypton in xenon for dark matter detectors.
- Investigated Kr and Xe excimer emission continua and assessed KrXe emission as a potential signature of trace krypton.
- Acquired experimental data through Linux-based digitiser control and implemented a binary random forest classifier to filter oscillatory background noise.

Oct. 2024 - Jun. 2025 **Gas-Phase Background Identification in the LUX-ZEPLIN Experiment**
MSci research project | University College London
Supervisors: Dr Aiham Al Musalhi and Dr Amy Cottle
- Developed an unsupervised Gaussian mixture model to identify gas-phase background events in liquid-xenon time projection chamber (TPC) data.
- Analysed discriminating features with boosted decision trees and refined the Gaussian mixture model using physics-informed weighting.
- Tagged 49% of background events in the WIMP region of interest as gas-phase backgrounds while retaining approximately 97% of ER and NR calibration events.
- Estimated that gas-background rejection could recover approximately 6.7% of the fiducial detector mass.

ACADEMIC PROJECTS

Beauty Meson Decay Analysis
- Applied particle-identification cuts, mass-hypothesis vetoes and a boosted decision tree to suppress backgrounds in a supplied LHCb decay sample.
- Extracted signal yields with an extended maximum-likelihood mass fit and used sWeights to study background-subtracted invariant-mass distributions.

Symmetry-Constrained Neural Networks
- Constructed scalar and vector neural architectures with built-in Lorentz symmetry and trained them to approximate supplied field-representation networks.
- Evaluated equivariance errors and numerically tested Lorentz and global SU(2) invariance of a constructed Lagrangian density.

Weak-Lensing Inference
- Compared galaxy-shape estimators on simulated images and constructed stacked tangential-shear profiles with shape-noise uncertainty estimates.
- Inferred halo mass and source redshift using a supplied NFW model, examining their posterior degeneracy and the effect of fixing redshift.

Bayesian Change-Point Modelling
- Inferred changes in historical coal-mining accident rates using Poisson change-point models, MCMC and nested sampling; compared models using Bayesian evidence.
- Completed and applied a supplied reversible-jump MCMC framework to estimate change-point counts and model-averaged rates with uncertainty intervals.

Neural Regression Application
- Built a configurable TensorFlow/Keras package for five-dimensional regression, with input validation, feature standardisation and separate training, validation and test datasets.
- Implemented training and prediction APIs with FastAPI and a Next.js interface; added pytest tests, Sphinx documentation and Docker Compose configuration.

Normalising-Flow Density Estimation
- Implemented an affine-coupling normalising flow in PyTorch, verifying invertibility and analytic Jacobians and controlling overfitting with early stopping.
- Studied transformed density families and analysed the computational cost of likelihood evaluation.

Image Classification and Metric Learning
- Tuned PyTorch image classifiers with Optuna and assessed training reproducibility and performance differences between supplied digit images and MNIST.
- Implemented triplet-loss embeddings for CIFAR-10 and compared learned representations with t-SNE visualisations.

Photon-Detector Calibration
- Compared individual and simultaneous likelihood fits to photon-detector calibration data and assessed parameter uncertainty and bias using bootstrap resampling.

CNN-Based Classification of Supernova Neutrino Events in LArTPC Images
- Trained CNNs to distinguish supernova-neutrino events from background in liquid-argon TPC images, generating and overlaying simulated electronic and radiological noise.
- Achieved test accuracy above 80% under the studied electronic-noise conditions and analysed classification failures caused by elongated radiological backgrounds.

SKILLS

- **Programming and data analysis:** Python; NumPy, SciPy, pandas, Matplotlib and Jupyter; data preprocessing, feature engineering and visualisation.
- **Statistical methods:** Maximum-likelihood and least-squares fitting, bootstrap uncertainty estimation, Bayesian inference, MCMC, nested sampling and model comparison.
- **Machine learning:** PyTorch, TensorFlow/Keras and scikit-learn; boosted decision trees, random forests, Gaussian mixture models, CNNs, Transformers, symmetry-constrained networks and normalising flows; Optuna hyperparameter optimisation, feature ablation and model evaluation.
- **Scientific data analysis:** ROOT data with uproot and Awkward Array; invariant-mass fitting, sWeights, GalSim and weak-lensing inference.
- **Research computing and software:** Git, Linux/SSH, Slurm GPU workflows, Python packaging, pytest and Sphinx; FastAPI, Next.js and Docker Compose project experience.
- **Laboratory experience:** Gas-handling and vacuum instrumentation; CCD cameras, monochromators, photomultiplier tubes and digitiser-based data acquisition.

EXTRACURRICULAR EXPERIENCE

- **Music:** DipABRSM in piano; ABRSM Grade 8 in violin and viola; seven years of orchestral experience.
- **Digital content creation:** Built and managed a channel covering gaming, music and digital art, attracting over 5,000 followers and generating over £2,000 in revenue.